

BikeFlow Console

A Micro-Mobility Bicycle Share Tracking and Management Dashboard

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Live Demo: <https://bike-flow-console.vercel.app/>

Problem Statement

BikeFlow Console is a micro-mobility bicycle share tracking and management system designed to monitor and manage bike stations across Mumbai city.

The system addresses the following real-world challenges:

1. Dock Availability Monitoring

Bike sharing stations often lack real-time visibility of Available bikes and empty docks, making it difficult for operators to manage fleet distribution.

2. Rental Transaction Management

Errors in bike rental transactions need a safe and reliable Undo mechanism to prevent data inconsistency and ensure accurate records.

3. Repair Queue Management

Bikes requiring maintenance need to be tracked and prioritized systematically to ensure timely repairs and minimize downtime.

4. Station Vacancy Analysis

Operators need a quick way to identify which stations
Have the most empty slots to plan efficient bike
Redistribution.

5. City Map

VisualizationA centralized map is needed to visualize all
Station locations across the city for better fleet management
and planning.

6.Truck Load Optimization

Maintenance trucks need an efficient system to assign and
manage bikes for redistribution across stations.

7. GPS Tracking (Planned)

Real-time GPS tracking of each bike requires IoT
hardware integration which is planned for future
implementation.

8. Safe Route Planning (Planned)

An intelligent route planning system requires
real-time GPS and IoT sensor data which is planned
for future implementation.

Tech Stack

The following technologies were used to build BikeFlow Console:

1. React.js

Frontend JavaScript library used for building the user interface through reusable components.

2. Vite

Modern build tool used for fast development server and optimized production builds.

3. React Leaflet + OpenStreetMap

Free and open source mapping library used for displaying interactive Mumbai city map with station markers. No API key required.

4. React Icons

Professional icon library used for sidebar navigation icons and UI elements.

5. JavaScript ES6

Core programming language used for implementing data structures including Stack and Queue logic.

6. CSS3

Used for styling all components with a clean and professional dashboard layout.

Component Architecture

BikeFlow Console follows a component based architecture where each feature is built as an independent React component.

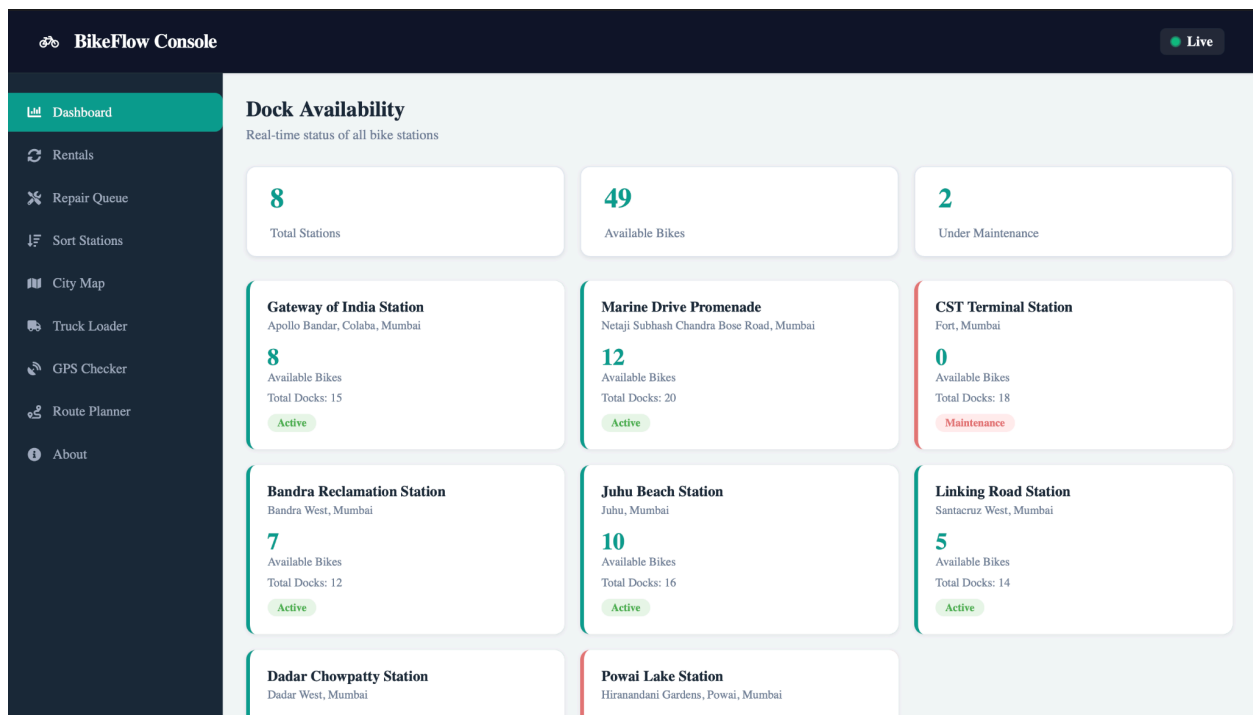
```
src/
├── components/
│   ├── Navbar.jsx      → Top navigation bar
│   ├── Sidebar.jsx     → Left navigation menu
│   ├── DockAvailability → Feature 1
│   ├── RentalUndo.jsx  → Feature 2
│   ├── RepairQueue.jsx → Feature 3
│   ├── EmptySlotSorter → Feature 4
│   ├── CityMap.jsx     → Feature 5
│   ├── TruckLoader.jsx → Feature 6
│   ├── GPSChecker.jsx  → Feature 7 (Placeholder)
│   ├── RoutePlanner.jsx → Feature 8 (Placeholder)
│   ├── Toast.jsx       → Notification component
│   └── About.jsx       → Project info page
├── data/
│   └── dummyData.js    → Simulated fleet data
├── styles/
│   └── main.css        → Global stylesheet
├── App.jsx            → Root component
└── main.jsx           → Entry point
```

Data Flow:

dummyData.js → App.jsx → Components → UI

Screenshots

1. Dock Availability Dashboard



2. Rental Management

4. Sort Stations

 BikeFlow Console

Live

Dashboard

Rentals

Repair Queue

Sort Stations

City Map

Truck Loader

GPS Checker

Route Planner

About

System Online

8 Stations - 49 Bikes

Empty Slot Sorter

Stations ranked by most vacant spots

Stations by Vacancy

Rank	Station Name	Location	Total Docks	Available Bikes	Empty Slots
1	CST Terminal Station	Fort, Mumbai	18	0	18
2	Powai Lake Station	Hiranandani Gardens, Powai, Mumbai	15	3	12
3	Linking Road Station	Santacruz West, Mumbai	14	5	9
4	Marine Drive Promenade	Netaji Subhash Chandra Bose Road, Mumbai	20	12	8
5	Gateway of India Station	Apollo Bandar, Colaba, Mumbai	15	8	7
6	Juhu Beach Station	Juhu, Mumbai	16	10	6
7	Dadar Chowpatty Station	Dadar West, Mumbai	10	4	6
8	Bandra Reclamation Station	Bandra West, Mumbai	12	7	5

5. City Map Hub

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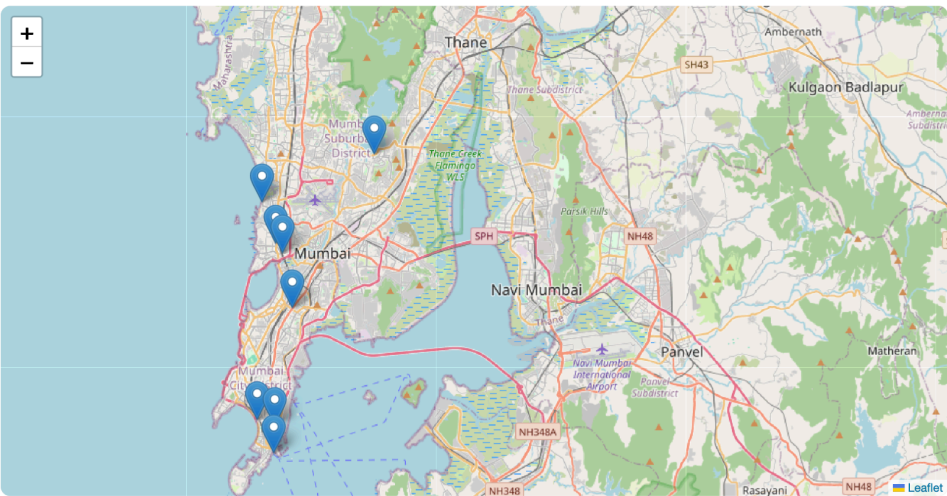
System Online

8 Stations - 49 Bikes

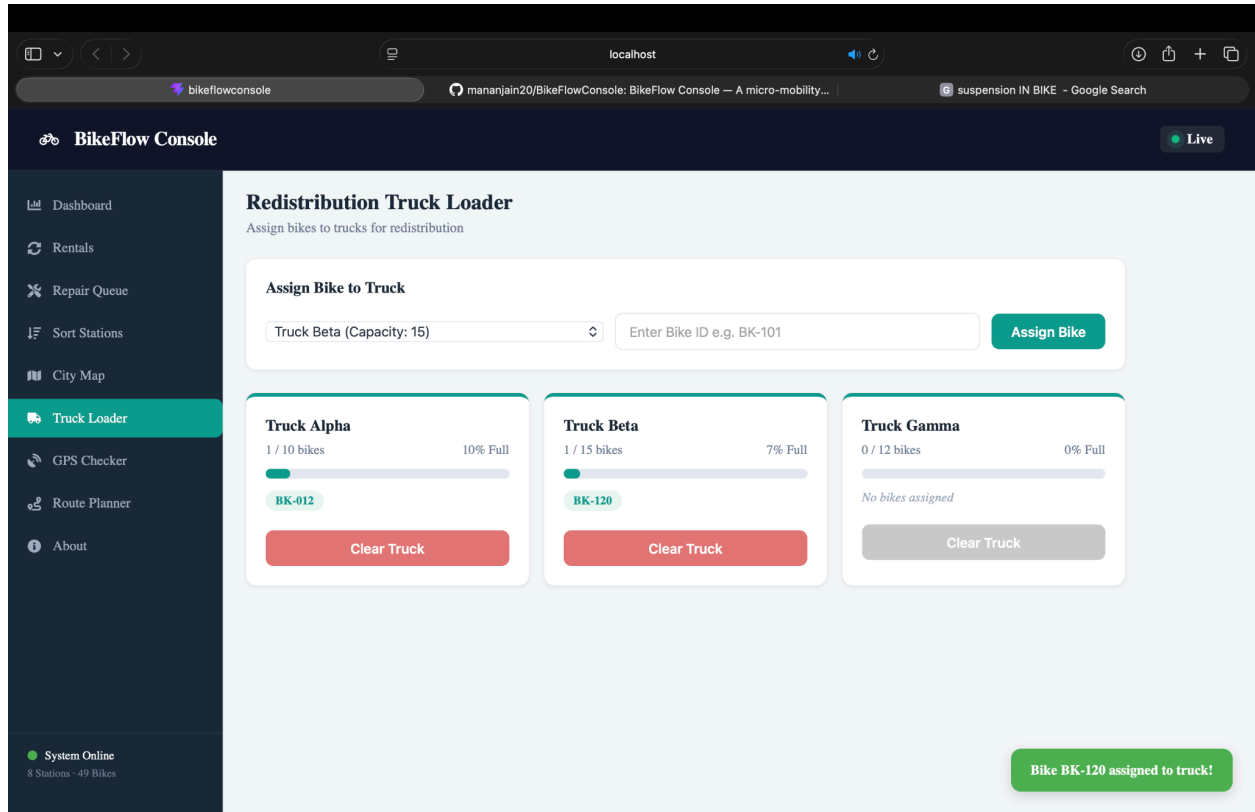
City Map Hub

All bike stations across Mumbai

Showing 8 active stations across the network



6. Truck Loader



7. GPS Checker

 BikeFlow Console

● Live

Dashboard

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8 Stations · 49 Bikes

GPS ID Checker

Verify bike tracking chip against fleet



Hardware Integration Required

GPS Tracking Module

This feature requires IoT hardware integration including GPS tracking chips installed on each bike and a real-time data receiver. This module has been designed and is ready for future hardware integration.

Why is this not implemented?

- IoT hardware integration not covered in curriculum
- Requires physical GPS chips on each bike
- Needs real-time data streaming infrastructure
- Planned for future production version

8. Route Planner

 BikeFlow Console

● Live

Dashboard

Rentals

Repair Queue

Sort Stations

City Map

Truck Loader

GPS Checker

Route Planner

About

● System Online

8 Stations · 49 Bikes

Safe Route Planner

Find safest routes avoiding dangerous areas



Hardware Integration Required

Route Planning Module

This feature requires real-time GPS data and IoT sensor integration to detect road conditions and avoid dangerous areas. The UI is designed and ready for future implementation.

Why is this not implemented?

- Requires real-time GPS and IoT sensors
- Dijkstra algorithm needs live road condition data
- Not covered in current curriculum
- Planned for future production version

Conclusion

BikeFlow Console successfully demonstrates a real-world micro-mobility bicycle share tracking and management system built using React.js.

The project implements core computer science concepts including:

- Stack data structure for rental undo operations
- Queue data structure for repair management
- Array sorting for station vacancy analysis
- Greedy logic for truck load optimization
- Interactive mapping using React Leaflet

All 6 core features are fully functional and demonstrated using simulated fleet data representing real Mumbai city bike stations. The remaining 2 features — GPS ID Checker and Safe Route Planner — have been thoughtfully designed with complete UI and are ready for future IoT hardware integration once the required infrastructure becomes available.

This project demonstrates not just frontend development skills but also the ability to translate real-world problems into working software solutions. BikeFlow Console is designed as a scalable prototype that can be extended into

a fully production-ready application in the future.

Project Links

GitHub Repository:

<https://github.com/mananjain20/BikeFlowConsole>

Live Demo:

bike-flow-console-jkz499xf1-mananjain.vercel.app